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Education & Career Success

1) Abstract

The study explores the factors that contribute to career success using the Education & Career Success dataset by Adil Shamim from Kaggle. The dataset obtained includes information on student demographics, academic performance, practical experience, skills, and career outcomes. Our research question targets to investigate the main contributing factors to higher starting salaries and overall job satisfaction after graduation.

To examine the relationships between academic track, academic performance, practical experience, salary, job offers, work-life balance, and career satisfaction, Exploratory Data Analysis (EDA) was conducted. The results of the analysis imply that STEM graduates receive the highest starting salaries, followed by ABM, HUMSS, and then A&D graduates. The results also suggest that practical experience has a stronger relationship with job offers than only academic performance. Additionally, higher starting salaries were associated with greater career satisfaction, while work-life balance showed a more complex relationship with both salary and satisfaction.

Overall, the findings highlight the importance of practical experience and skill development alongside academic achievement in contributing to career success and improved employment outcomes.

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2) Dataset Description

The "Education & Career Success" dataset is collected by Adil Shamim found on Kaggle. The dataset includes data collected from 400 anonymized student records. The purpose of this dataset is to explore how students' academic achievements, skills, and extracurricular activities translate into real-world career outcomes. The key features are:

- `Student Demographics`
  - **Student_ID** — Unique alphanumeric identifier for each student (ensures privacy while enabling record linkage)
  - **Age** — Student's age in whole years at graduation or data collection *(range: 18–30)*
  - **Gender** — Self-reported gender identity: `"Male"`, `"Female"`, or `"Other"`
- `Academic Performance`
  - **High_School_GPA** — Cumulative high school GPA *(scale: 2.0–4.0)*
  - **SAT_Score** — Total SAT score *(range: 900–1600)*
  - **University_GPA** — Cumulative undergraduate GPA *(scale: 2.0–4.0)*
  - **Field_of_Study** — Major or academic discipline (e.g., `"Computer Science"`, `"Medicine"`, `"Business"`)
- `Skills & Extracurriculars`
  - **Internships_Completed** — Number of formal internships completed during college *(range: 0–4)*
  - **Projects_Completed** — Count of significant academic or personal projects finished *(range: 0–9)*
  - **Certifications** — Number of industry-recognized certificates earned beyond coursework *(range: 0–5)*
  - **Soft_Skills_Score** — Instructor- or peer-rated measure of communication, teamwork, and leadership *(scale: 1–10)*
  - **Networking_Score** — Quantitative assessment of professional networking activity *(scale: 1–10)*
- `Career Outcomes`
  - **Job_Offers** — Total distinct full-time job offers received within six months of graduation *(range: 0–5)*
  - **Starting_Salary** — Annual base salary (USD) for first full-time position *(range: $25,000–150,000$)*
  - **Career_Satisfaction** — Self-reported satisfaction with first job *(1 = very dissatisfied, 10 = very satisfied)*
  - **Years_to_Promotion** — Years from hire date to next-level promotion *(range: 1–5)*
  - **Current_Job_Level** — Career stage at data collection: `"Entry"`, `"Mid"`, `"Senior"`, or `"Executive"`
  - **Work_Life_Balance** — Self-assessed work-life balance *(1 = poor, 10 = excellent)*
  - **Entrepreneurship** — Whether the individual founded or co-founded a business (`"Yes"` or `"No"`)
- `Possible Limitations`
  - The dataset does not specify the geographic location or demographic composition of the participants, making it difficult to assess its representativeness. Additionally, variables such as Networking_Score lack a documented measurement methodology, which may affect the interpretation of the results.

Link to dataset: **Education & Career Success**

In [210…

```python
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import sys
from scipy.stats import shapiro, levene, ttest_ind, f_oneway, chi2_contingency


try:
    # Load dataset
    # If in colab, prompt file upload
    # Else, find dataset in the same folder
    try:
        from google.colab import files
        IN_COLAB = True
    except ImportError:
        IN_COLAB = False

    if IN_COLAB:
        from google.colab import files
        uploaded = files.upload()
        filename = next(iter(uploaded))
    else:
        filename = 'education_career_success.csv'

    ecs_df = pd.read_csv(
        filename,
        index_col="Student_ID"
    )

    print(f"Successfully loaded {filename}")

except FileNotFoundError:
    print("File not found.", file=sys.stderr)

except pd.errors.EmptyDataError:
    print("CSV is empty.", file=sys.stderr)

except pd.errors.ParserError:
    print("Invalid CSV format.", file=sys.stderr)

except Exception as e:
    print(f"Error: {e}", file=sys.stderr)

else:
    display(ecs_df.head())
```

```
Successfully loaded education_career_success.csv
```

Student_ID	Age	Gender	High_School_GPA	SAT_Score	University_GPA	Field_of_Study	Internships_Completed	Projects_Completed	Certifications	Soft_Skills_Score	Networking_Score	Job_Offers	Starting_Salary	Career_Satisfaction	Years_to_Promotion	Current_Job_Level
S001	22	Male	3.80	1450	3.60	Computer Science	3	7	2	8	7	3	85000	8	2	Mid
S002	24	Female	3.60	1380	3.40	Business	2	5	3	7	6	2	65000	7	3	Mid
S003	21	Male	3.90	1520	3.80	Engineering	4	9	4	9	8	4	120000	9	1	Senior
S004	23	Female	3.50	1300	3.20	Psychology	1	3	1	6	5	1	48000	6	4	Entry
S005	25	Male	3.70	1420	3.50	Medicine	2	6	2	8	7	3	95000	8	2	Mid

In [211…

```python
# Overview description of the table
ecs_df.describe(percentiles=None, include=None, exclude=None)
```

Out[211…

	Age	High_School_GPA	SAT_Score	University_GPA	Internships_Completed	Projects_Completed	Certifications	Soft_Skills_Score	Networking_Score	Job_Offers	Starting_Salary	Career_Satisfaction	Years_to_Promotion	Work_Life_Balance
count	400.00	400.00	400.00	400.00	400.00	400.00	400.00	400.00	400.00	400.00	400.00	400.00	400.00	400.00
mean	23.00	3.62	1388.60	3.44	2.50	6.24	2.62	7.93	6.93	2.74	87562.50	7.79	2.92	6.41
std	1.42	0.19	114.31	0.29	1.06	1.90	1.27	1.31	1.32	1.36	29685.42	1.37	1.34	0.93
min	21.00	3.20	1160.00	2.80	0.00	2.00	0.00	5.00	4.00	0.00	31000.00	5.00	1.00	5.00
25%	22.00	3.50	1300.00	3.20	2.00	5.00	2.00	7.00	6.00	2.00	68000.00	7.00	2.00	6.00

50%	23.00	3.60	1380.00	3.45	3.00	6.00	3.00	8.00	7.00	3.00	84000.00	8.00	3.00	6.00
75%	24.00	3.80	1480.00	3.70	3.00	8.00	3.00	9.00	8.00	4.00	105250.00	9.00	4.00	7.00
max	25.00	3.90	1580.00	4.00	4.00	9.00	5.00	10.00	9.00	5.00	152000.00	10.00	5.00	9.00

---

3) Data Cleaning and Preprocessing

For the data cleaning and preprocessing, the dataset we have obtained is already clean and well-structured. However, we performed some basic checks to ensure there are no missing values or inconsistencies in the data.

In [212…

```
ecs_df.isnull().sum()
```

Out[212…

```
Age                      0
Gender                   0
High_School_GPA          0
SAT_Score                0
University_GPA           0
Field_of_Study           0
Internships_Completed    0
Projects_Completed       0
Certifications           0
Soft_Skills_Score        0
Networking_Score         0
Job_Offers               0
Starting_Salary          0
Career_Satisfaction      0
Years_to_Promotion       0
Current_Job_Level        0
Work_Life_Balance        0
Entrepreneurship         0
dtype: int64
```

From the output above, we can conclude that there were no missing values in the dataset.

In [213…

```
ecs_df.duplicated().sum()
```

Out[213…

```
np.int64(232)
```

From the result, there seems to be 232 observed duplicates. To further inspect the duplicates:

In [214…

```
dup_rows = ecs_df[ecs_df.duplicated(keep=False)]
dup_rows.sort_values(list(ecs_df.columns)).head(20)
```

Out[214…

	Age	Gender	High_School_GPA	SAT_Score	University_GPA	Field_of_Study	Internships_Completed	Projects_Completed	Certifications	Soft_Skills_Score	Networking_Score	Job_Offers	Starting_Salary	Career_Satisfaction	Years_to_Promotion	Current_Job_Leve
Student_ID																
S154	21	Female	3.40	1260	3.10	Psychology	1	4	1	6	5	1	56000	6	5	Entry
S224	21	Female	3.40	1260	3.10	Psychology	1	4	1	6	5	1	56000	6	5	Entry
S284	21	Female	3.40	1260	3.10	Psychology	1	4	1	6	5	1	56000	6	5	Entry
S354	21	Female	3.40	1260	3.10	Psychology	1	4	1	6	5	1	56000	6	5	Entry
S374	21	Female	3.40	1260	3.10	Psychology	1	4	1	6	5	1	56000	6	5	Entry
S394	21	Female	3.40	1260	3.10	Psychology	1	4	1	6	5	1	56000	6	5	Entry
S114	21	Female	3.50	1300	3.20	Marketing	2	5	2	7	6	2	64000	7	4	Mic
S184	21	Female	3.50	1300	3.20	Marketing	2	5	2	7	6	2	64000	7	4	Mic
S244	21	Female	3.50	1300	3.20	Marketing	2	5	2	7	6	2	64000	7	4	Mic
S314	21	Female	3.50	1300	3.20	Marketing	2	5	2	7	6	2	64000	7	4	Mic
S134	21	Female	3.50	1320	3.30	Engineering	2	5	2	7	6	2	72000	7	4	Mic
S204	21	Female	3.50	1320	3.30	Engineering	2	5	2	7	6	2	72000	7	4	Mic
S264	21	Female	3.50	1320	3.30	Engineering	2	5	2	7	6	2	72000	7	4	Mic
S334	21	Female	3.50	1320	3.30	Engineering	2	5	2	7	6	2	72000	7	4	Mic
S124	21	Female	3.70	1440	3.50	Marketing	3	7	3	8	7	3	72000	8	3	Mic
S194	21	Female	3.70	1440	3.50	Marketing	3	7	3	8	7	3	72000	8	3	Mic
S254	21	Female	3.70	1440	3.50	Marketing	3	7	3	8	7	3	72000	8	3	Mic
S324	21	Female	3.70	1440	3.50	Marketing	3	7	3	8	7	3	72000	8	3	Mic
S164	21	Female	3.70	1460	3.50	Psychology	3	7	3	8	7	3	90000	8	2	Mic
S294	21	Female	3.70	1460	3.50	Psychology	3	7	3	8	7	3	90000	8	2	Mic

Upon inspection, these duplicates corresponded to repeated student profiles with identical attributes but different Student_ID values. Since the dataset appears to be synthetically generated, these repeated profiles were treated as part of the dataset design rather than data-entry errors.

Additionally, we created tracks for fields of study, STEM, ABM, HUMSS and A&D as a form of feature engineering.

Create a new column for **"Track"**

In [215…

```
track_map = {
    'Computer Science': 'STEM',
    'Engineering': 'STEM',
    'Medicine': 'STEM',
    'Nursing': 'STEM',

    'Business': 'ABM',
    'Finance': 'ABM',
    'Marketing': 'ABM',

    'Law': 'HUMSS',
    'Psychology': 'HUMSS',
    'Education': 'HUMSS',

    'Arts': 'A&D'
}

ecs_df['Track'] = ecs_df['Field_of_Study'].map(track_map)

#counts how many students are under each track
ecs_df['Track'].value_counts()
```

Out[215…

```
Track
STEM     145
ABM      121
HUMSS    102
A&D       32
Name: count, dtype: int64
```

---

4) Exploratory Data Analysis (EDA)

The given dataset contains various columns that can be analyzed to understand the relationships between education and career success. We have our main research question:

"What are the main contributing factors for a high salary and overall job satisfaction?"

As part of the exploratory data analysis, and to answer our main research question, the following questions were formed:

EDA Question 1

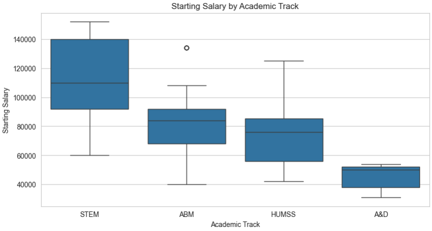
How does average starting salary differ among academic tracks;

1. STEM (Science, Technology, Engineering, and Mathematics)
2. ABM (Accountancy, Business, and Management)
3. HUMSS (Humanities and Social Sciences)
4. A&D (Arts and Design)

In [216…

```
plt.figure(figsize=(10,5))
```

```
sns.boxplot(
    data=ecs_df,
    x='Track',
    y='Starting_Salary'
)

plt.title('Starting Salary by Academic Track')
plt.xlabel('Academic Track')
plt.ylabel('Starting Salary')

plt.show()
```



EDA Question 1 Findings

The plot helps visualize the distribution of starting salaries across academic tracks. STEM graduates tend to receive the highest starting salaries, with a median salary of approximately 110,000 USD and an upper range reaching around 150,000 USD. This suggests that careers associated with STEM fields generally offer higher initial compensation than other tracks.

ABM has the second-highest median salary at approximately 84,000 USD. Most salaries are concentrated between 70,000 USD and 90,000 USD, indicating a relatively consistent salary distribution compared to STEM.

HUMSS ranks third among the four tracks, with a median salary of approximately 76,000 USD and a wider salary range, indicating greater variability in career outcomes.

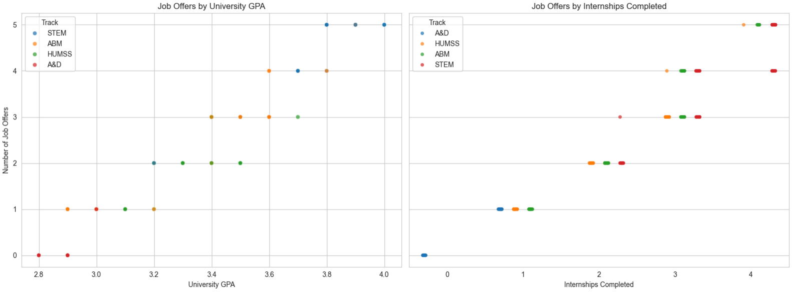
A&D graduates have the lowest median starting salary at approximately 50,000 USD, suggesting that careers in arts and design generally offer lower initial compensation compared to the other academic tracks.

EDA Question 2

What has a bigger impact on the number of job offers received: academic performance (University_GPA) or practical experience (Internships_Completed)? How does this relationship behave across different academic tracks?

In [217…

```
import matplotlib.pyplot as plt
import seaborn as sns

# Set up a side-by-side plot to compare GPA vs Internships against Job Offers
fig, axes = plt.subplots(1, 2, figsize=(16, 6), sharey=True)

# Plot 1: University GPA vs Job Offers
sns.scatterplot(
    data=ecs_df,
    x='University_GPA',
    y='Job_Offers',
    hue='Track',
    alpha=0.7,
    ax=axes[0]
)
axes[0].set_title('Job Offers by University GPA')
axes[0].set_xlabel('University GPA')
axes[0].set_ylabel('Number of Job Offers')

# Plot 2: Internships Completed vs Job Offers
# Using a stripplot/boxplot combination since Internships is a discrete count
sns.stripplot(
    data=ecs_df,
    x='Internships_Completed',
    y='Job_Offers',
    hue='Track',
    dodge=True,
    alpha=0.7,
    ax=axes[1]
)
axes[1].set_title('Job Offers by Internships Completed')
axes[1].set_xlabel('Internships Completed')
axes[1].set_ylabel('') # Shared y-axis

plt.tight_layout()
plt.show()
```



EDA Question 2 Findings

The plots provide a comparative look at how academic performance versus real-world experience influences a student's job market competitiveness:

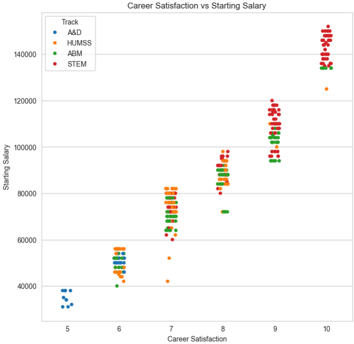
1. **The Internship Advantage:** Looking at the second plot, there is a clear, stepwise upward trend between `Internships_Completed` and `Job_Offers`. Students with 3 to 4 internships consistently land in the upper tier of job offers (3 to 5 offers), regardless of their track.
2. **The GPA Threshold:** While a higher `University_GPA` generally correlates with more job offers, the distribution is much more scattered. A high GPA alone does not guarantee maximum job offers if practical experience is missing.
3. **Track-Specific Behaviors:** STEM and ABM students leverage both high GPAs and high internship counts effectively to maximize their offer pools, whereas A&D and HUMSS tracks show a heavier reliance on practical indicators over perfect GPA scores to secure multiple offers.

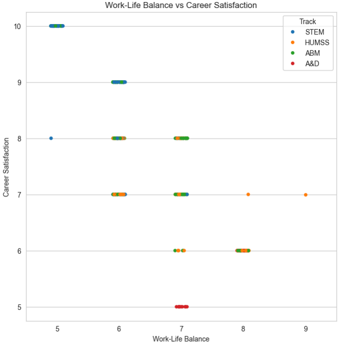
EDA Question 3

How do the relationships between salary, job satisfaction, and work-life balance look like?

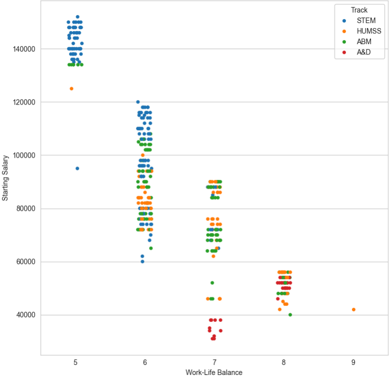
In [218…

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8, 8))
sns.stripplot(
    data=ecs_df,
    x='Career_Satisfaction',
    y='Starting_Salary',
    hue='Track'
)
plt.title('Career Satisfaction vs Starting Salary')
plt.xlabel('Career Satisfaction')
plt.ylabel('Starting Salary')

plt.figure(figsize=(8, 8))
sns.stripplot(
    data=ecs_df,
    x='Work_Life_Balance',
    y='Career_Satisfaction',
    hue='Track'
)
plt.title('Work-Life Balance vs Career Satisfaction')
plt.xlabel('Work-Life Balance')
plt.ylabel('Career Satisfaction')

plt.figure(figsize=(8, 8))
sns.stripplot(
    data=ecs_df,
    x='Work_Life_Balance',
    y='Starting_Salary',
    hue='Track'
)
plt.title('Work-Life Balance vs Starting Salary')
plt.xlabel('Work-Life Balance')
plt.ylabel('Starting Salary')

plt.tight_layout()
plt.show()
```





Work-Life Balance vs Starting Salary



EDA Question 3 Findings

The plots help visualize the correlations between starting salary, work-life balance, and career satisfaction.

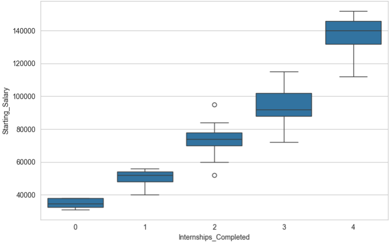
1. The first plot indicates a positive correlation between starting salary and career satisfaction, with STEM and ABM dominating the upper half of the graph while HUMSS and A&D appearing on the lower half.
2. The second plot indicates a negative correlation between Career Satisfaction and Work-Life Balance.
3. The third plot indicates a negative correlation between Work-Life Balance and Starting Salary.

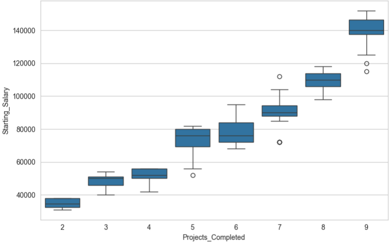
EDA Question 4

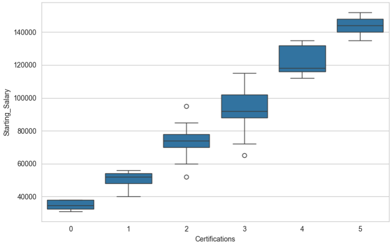
How do certifications, projects, and internships influence starting salary?

In [219…

```
import matplotlib.pyplot as plt
import seaborn as sns

fig, axes = plt.subplots(3, 1, figsize=(8, 15))

sns.boxplot(
    data=ecs_df,
    x='Internships_Completed',
    y='Starting_Salary',
    ax=axes[0]
)

sns.boxplot(
    data=ecs_df,
    x='Projects_Completed',
    y='Starting_Salary',
    ax=axes[1]
)

sns.boxplot(
    data=ecs_df,
    x='Certifications',
    y='Starting_Salary',
    ax=axes[2]
)

plt.tight_layout()
plt.show()
```







To check the median:

In [220…

```
print(ecs_df.groupby("Certifications")["Starting_Salary"].median())

print(ecs_df.groupby("Internships_Completed")["Starting_Salary"].median())

print(ecs_df.groupby("Projects_Completed")["Starting_Salary"].median())
```

```
Certifications
0    34500.00
1    52000.00
2    74000.00
3    92000.00
4   118000.00
5   144000.00
Name: Starting_Salary, dtype: float64
Internships_Completed
0    34500.00
1    52000.00
2    74000.00
3    92000.00
4   140000.00
Name: Starting_Salary, dtype: float64
Projects_Completed
2    34500.00
3    50000.00
4    52000.00
5    76000.00
6    76000.00
7    90000.00
8   110000.00
9   140000.00
Name: Starting_Salary, dtype: float64
```

EDA Question 4 Findings

From the plots above, we can see that there is a very strong positive relationship between practical experience indicators and starting salary. All three plots above show a clear upward trend, showing that students with more internships, completed projects, and certifications generally receive higher starting salaries.

From the three variables, `Certifications` appear to have the strongest relationship with salary, with a $109,500 increase. The same relationship is observed for `Internships_Completed`, where students who have more internships tend to have a higher starting salary. Lastly, it is also the same for `Projects_Completed`, where students who completed a greater number of projects also generally achieved a higher salary, which suggests that hands-on experiences are valued by employers.

Overall, the plots above suggest that students who are active with real-world application of learnings, such as internships, projects, and certifications, tend to have better financial outcomes after graduation. This highlights the importance of practical experience alongside academic achievement in contributing to career success.

EDA Question 5

Does the Field of Study influence the number of Skills & Extracurriculars (internships, projects, etc.) of the students?

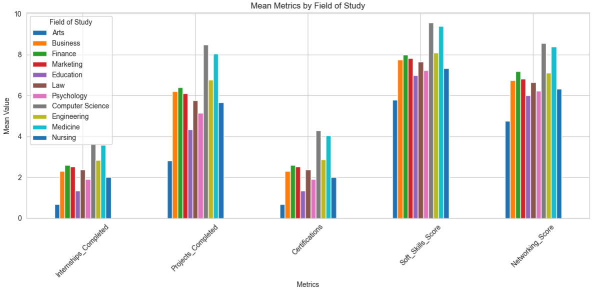
- What are the means of the columns under the skills & extracurriculars columns for each field of study?
- What are the top 5 from each column?

To answer EDA question 4, the Fields of Study were grouped by Track then Field_of_study then getting the means of columns under Skills & Extracurriculars. To visually plot, a grouped bar graph was used to compare each field under different skills & extracurriculars. The top 5 from each column was then retrieved then displayed.

In [221…

```
# Group the entries by Track then by Field_of_Study
# Get the columns under the Skills & Extracurriculars
eda4 = ecs_df.groupby(["Track","Field_of_Study"])[['Internships_Completed','Projects_Completed','Certifications','Soft_Skills_Score','Networking_Score']].mean()

# Set display for float data types to 2 decimal places
pd.set_option('display.float_format', lambda x: '%.2f' % x)

display(eda4)

eda4 = eda4.reset_index()

# Pivot so Field_of_Study becomes columns, then transpose
eda4_plot = eda4.set_index('Field_of_Study')[
    ['Internships_Completed','Projects_Completed','Certifications',
     'Soft_Skills_Score','Networking_Score']
].T

eda4_plot.plot(kind='bar', figsize=(12, 6))

plt.title('Mean Metrics by Field of Study')
plt.ylabel('Mean Value')
plt.xlabel('Metrics')
plt.xticks(rotation=45)
plt.legend(title='Field of Study')
plt.tight_layout()
plt.show()

# Get the top 5 from each column for easier analysis
eda4b = ecs_df.groupby("Field_of_Study")[['Internships_Completed','Projects_Completed','Certifications','Soft_Skills_Score','Networking_Score']].mean()
for col in eda4b.columns:
    print(f"\nTop 5 - {col}:")
    print(eda4b[col].nlargest(5))
```

		Internships_Completed	Projects_Completed	Certifications	Soft_Skills_Score	Networking_Score
Track	**Field_of_Study**					
A&D	**Arts**	0.69	2.81	0.69	5.78	4.75
ABM	**Business**	2.29	6.19	2.31	7.75	6.75
	Finance	2.60	6.40	2.60	8.00	7.20
	Marketing	2.52	6.11	2.52	7.82	6.82
HUMSS	**Education**	1.33	4.33	1.33	7.00	6.00
	Law	2.39	5.77	2.39	7.66	6.66
	Psychology	1.91	5.15	1.91	7.24	6.24
STEM	**Computer Science**	3.65	8.49	4.29	9.57	8.57
	Engineering	2.84	6.76	2.86	8.10	7.10
	Medicine	3.57	8.06	4.06	9.40	8.40
	Nursing	2.00	5.67	2.00	7.33	6.33



```
Top 5 - Internships_Completed:
Field_of_Study
Computer Science   3.65
Medicine           3.57
Engineering        2.84
Finance            2.60
Marketing          2.52
Name: Internships_Completed, dtype: float64

Top 5 - Projects_Completed:
Field_of_Study
Computer Science   8.49
Medicine           8.06
Engineering        6.76
Finance            6.40
Business           6.19
Name: Projects_Completed, dtype: float64

Top 5 - Certifications:
Field_of_Study
Computer Science   4.29
Medicine           4.06
Engineering        2.86
Finance            2.60
Marketing          2.52
Name: Certifications, dtype: float64

Top 5 - Soft_Skills_Score:
Field_of_Study
Computer Science   9.57
Medicine           9.40
Engineering        8.10
Finance            8.00
Marketing          7.82
Name: Soft_Skills_Score, dtype: float64

Top 5 - Networking_Score:
Field_of_Study
Computer Science   8.57
Medicine           8.40
Finance            7.20
Engineering        7.10
Marketing          6.82
Name: Networking_Score, dtype: float64
```

EDA Question 5 Findings

The plots help visualize the correlations between starting salary, work-life balance, and career satisfaction.

The plot gives a visual of the comparisons for each field of study within a certain skill or extracurricular, visually comparing each field. The top 5 from each column was then gathered for better analysis.

1. The first table shows the result of the grouping, showing the average across each field of study. From a quick read from the values, we can assume that fields under the STEM field have a higher averages across all skills & extracurriculars.
2. The plot visually compares the values against each other. While it may be difficult to visually digest, from a quick observation of the peak values show that a majority of the peaks come from STEM related fields supporting the previous observation.
3. The next series of tables show the Top 5 fields across each column. The following observations were made:

   a. All fields share the exact same Top 4, with an outlier within the Projects completed column where buisness beats out marketing. Other than the projects completed column, all the other fields share the same top 5 in the same order: Computer Science, Medicine, Finance, Engineering, and Marketing

   b. STEM fields appear to dominate the top with 3 of the 4 constantly on top of each of the column fields. The only field missing out the top 5 is Nursing. When looking at its means across each column we can see that

   c. The Arts field, the only field under A&D track, appears to consistenly rank the lowest across all fields. Observing the plot, the highest that this field scores is within the soft skills score with a mean of 5.78.
4. The same top 4 is observed with the same order accross all columns. The only outlier in the projects completed column. The two fields that differ for the 5th position are Marketing and Business fields which both fall under the ABM track. From this we can assume that for the Skills & Extracurricular columns are dominated by STEM related fields with ABM related fields a close second. HUMS related fields ranking third and A&D ranking last.

A concrete asnwer cannot be created from this data alone, thus the EDA question cannot be answered at this time. It is too early to say that the Fields of Study alone influence the Skills & Extracurricular columns. However, we can say that there is some relationship observed between the two.

5) Data Mining

K-Means Clustering (Feel free to edit)

In [222…

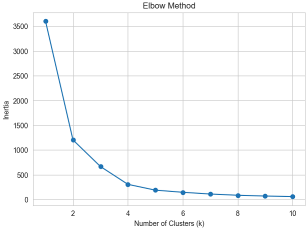
```
features = [
    "University_GPA",
    "Internships_Completed",
    "Projects_Completed",
    "Certifications",
    "Soft_Skills_Score",
    "Networking_Score",
    "Job_Offers",
    "Starting_Salary",
    "Career_Satisfaction"
]
```

In [223…

```
from sklearn.preprocessing import StandardScaler

X = ecs_df[features]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

In [224…

```
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

inertia = []

for k in range(1, 11):
    km = KMeans(n_clusters=k, random_state=42)
    km.fit(X_scaled)
    inertia.append(km.inertia_)

plt.figure(figsize=(7,5))
plt.plot(range(1,11), inertia, marker='o')
plt.xlabel("Number of Clusters (k)")
plt.ylabel("Inertia")
plt.title("Elbow Method")
plt.show()
```



In [225…

```
kmeans = KMeans(n_clusters=5, random_state=42)

ecs_df["Cluster"] = kmeans.fit_predict(X_scaled)
```

In [226…

```
ecs_df["Cluster"].value_counts().sort_index()
```

Out[226…

```
Cluster
0     85
1     79
2     58
3    107
4     71
Name: count, dtype: int64
```

In [227…

```
cluster_summary = ecs_df.groupby("Cluster")[features].mean()

display(cluster_summary)
```

	University_GPA	Internships_Completed	Projects_Completed	Certifications	Soft_Skills_Score	Networking_Score	Job_Offers	Starting_Salary	Career_Satisfaction
Cluster									
0	3.48	2.99	6.74	2.98	8.09	7.09	3.00	88152.94	8.00
1	3.02	0.87	3.37	0.87	6.09	5.08	0.87	48683.54	5.89
2	3.87	4.00	9.00	4.86	10.00	9.00	5.00	141793.10	10.00
3	3.33	2.00	5.45	2.01	7.34	6.35	2.00	73785.05	7.04
4	3.68	3.25	7.75	3.25	9.00	8.00	3.77	106577.46	9.00

In [228…

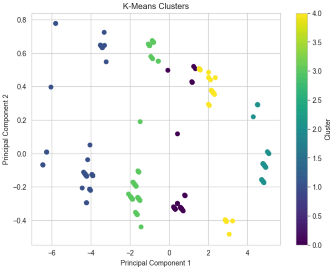
```
pd.crosstab(
    ecs_df["Cluster"],
    ecs_df["Track"]
)
```

Out[228…

Track	A&D	ABM	HUMSS	STEM
Cluster				
0	0	31	33	21
1	32	20	27	0
2	0	6	1	51
3	0	40	39	28
4	0	24	2	45

In [229…

```
from sklearn.decomposition import PCA

pca = PCA(n_components=2)

X_pca = pca.fit_transform(X_scaled)

plt.figure(figsize=(8,6))

plt.scatter(
    X_pca[:,0],
    X_pca[:,1],
    c=ecs_df["Cluster"],
    cmap="viridis"
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("K-Means Clusters")
plt.colorbar(label="Cluster")
plt.show()
```



`perform DATA MINING here`

`AT LEAST ONE (1)`

6) Statistical Inference

A. Is there a significant difference in starting salary among the four academic tracks?

This answers the question, is there a significant differenve in starting salary among the four academic tracks?

1. Variables

- Track
- Starting Salary

2. Hypotheses

- H_0: Mean salary is the same across all tracks.
- Ha: At least one track differs.

One-Way ANOVA

In [230…

```
#Seperate Groups
stem = ecs_df.loc[ecs_df["Track"]=="STEM","Starting_Salary"]
abm = ecs_df.loc[ecs_df["Track"]=="ABM","Starting_Salary"]
humss = ecs_df.loc[ecs_df["Track"]=="HUMSS","Starting_Salary"]
ad = ecs_df.loc[ecs_df["Track"]=="A&D","Starting_Salary"]
```

In [231…

```
#Test for Normality
for name, group in {
    "STEM": stem,
    "ABM": abm,
    "HUMSS": humss,
    "A&D": ad
}.items():

    stat, p = shapiro(group)

    print(name)
    print("Statistic:", stat)
    print("p-value:", p)
    print()
```

```
STEM
Statistic: 0.9268975511027331
p-value: 8.812328734754712e-07

ABM
Statistic: 0.9588784710004226
p-value: 0.0009716653417938198

HUMSS
Statistic: 0.9398861492261023
p-value: 0.0001624786047891139

A&D
Statistic: 0.7926730487109013
p-value: 2.9773607827950638e-05
```

In [232…

```
#Equal Test Variance
stat, p = levene(stem, abm, humss, ad)

print(stat)
print(p)
```

```
23.71191551872145
3.9155796141466445e-14
```

In [233…

```
stat, p = f_oneway(stem, abm, humss, ad)

print("F-statistic:", stat)
print("p-value:", p)
```

```
F-statistic: 116.07183373420608
p-value: 6.098472543721593e-54
```

Findings from the ANOVA

P value is less than SIgnificance level (0.05), should reeject null hypothesis.

This suggest that the mean starting salary differs significantly among the four academic tracks. There is strong evidence that academic track has association with differences in starting salaries.

Tukey HSD Post-Hoc Test

In [234…

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd

tukey = pairwise_tukeyhsd(
    endog=ecs_df["Starting_Salary"],
    groups=ecs_df["Track"],
    alpha=0.05
)

print(tukey)
```

```
    Multiple Comparison of Means - Tukey HSD, FWER=0.05
=============================================================
group1 group2  meandiff  p-adj     lower      upper    reject
-------------------------------------------------------------
   A&D    ABM 35195.5062    0.0  24048.1101 46342.9023   True
   A&D  HUMSS 27523.8971    0.0  16161.4279 38886.3662   True
   A&D   STEM 65664.8707    0.0  54712.1242 76617.6172   True
   ABM  HUMSS -7671.6091 0.0443 -15209.5904  -133.6279   True
   ABM   STEM 30469.3645    0.0  23564.4306 37374.2984   True
 HUMSS   STEM 38140.9736    0.0  30893.9483  45387.999   True
-------------------------------------------------------------
```

Findings from Tukey HSD

The Tukey HSD post-hoc test showed that all pairwise comparisons between the academic tracks were statistically significant ($p < 0.05$). STEM graduates had significantly higher mean starting salaries than ABM, HUMSS, and A&D graduates. Similarly, ABM graduates earned significantly higher salaries than HUMSS and A&D graduates, while HUMSS graduates also earned significantly higher salaries than A&D graduates. These results confirm the findings from the ANOVA and EDA, establishing a clear ranking of mean starting salaries:

STEM > ABM > HUMSS > A&D.

A.2. Is there a significant difference in career satisfaction among the four academic tracks?

Null Hypothesis (H_0): There is no significant difference in the mean career satisfaction among the four academic tracks.

Alternative Hypothesis (Ha): At least one academic track has a significantly different mean career satisfaction.

One-Way ANOVA

In [235…

```
stem = ecs_df.loc[ecs_df["Track"]=="STEM","Career_Satisfaction"]
abm = ecs_df.loc[ecs_df["Track"]=="ABM","Career_Satisfaction"]
humss = ecs_df.loc[ecs_df["Track"]=="HUMSS","Career_Satisfaction"]
ad = ecs_df.loc[ecs_df["Track"]=="A&D","Career_Satisfaction"]

#perform one way anova
F, p = f_oneway(stem, abm, humss, ad)

#print results
print("F-statistic:", F)
print("p-value:", p)
```

```
F-statistic: 109.4067172077476
p-value: 1.3192702074052227e-51
```

Findings

The One-Way ANOVA was conducted to determine whether the mean career satisfaction differed among the four academic tracks. The analysis produced an F-statistic of 109.41 and a p-value of 1.32×10^{-51}, which is significantly lower than the chosen significance level of 0.05. Therefore, the null hypothesis was rejected, indicating that there is a statistically significant difference in mean career satisfaction among the academic tracks. These results suggest that a graduate's academic track is associated with differences in overall career satisfaction after graduation.

Tukey HSD Post-Hoc Test

In [236…

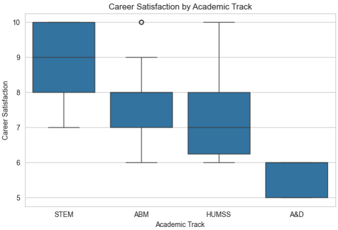
```
tukey = pairwise_tukeyhsd(
    endog=ecs_df["Career_Satisfaction"],
    groups=ecs_df["Track"],
    alpha=0.05
)

print(tukey)
```

```
Multiple Comparison of Means - Tukey HSD, FWER=0.05
====================================================
group1 group2 meandiff p-adj   lower   upper  reject
----------------------------------------------------
```

```
  A&D    ABM   1.9819    0.0  1.4594  2.5044   True
  A&D  HUMSS   1.4498    0.0  0.9172  1.9824   True
  A&D   STEM   3.1332    0.0  2.6198  3.6466   True
  ABM  HUMSS  -0.5322 0.0007 -0.8855 -0.1788   True
  ABM   STEM   1.1513    0.0  0.8276  1.4749   True
HUMSS   STEM   1.6834    0.0  1.3437  2.0231   True
----------------------------------------------------
```

In [237…

```python
#this is the boxplot same with EDA 1 but with career satisfaction
plt.figure(figsize=(8,5))

sns.boxplot(
    data=ecs_df,
    x="Track",
    y="Career_Satisfaction"
)

plt.title("Career Satisfaction by Academic Track")
plt.xlabel("Academic Track")
plt.ylabel("Career Satisfaction")

plt.show()
```



The Tukey HSD post-hoc test revealed that all pairwise comparisons among the academic tracks were statistically significant (p < 0.05). STEM graduates reported significantly higher mean career satisfaction than ABM, HUMSS, and A&D graduates. Likewise, ABM graduates had significantly higher career satisfaction than HUMSS and A&D graduates, while HUMSS graduates also reported significantly higher career satisfaction than A&D graduates. These findings establish a consistent ranking of mean career satisfaction across the academic tracks: STEM > ABM > HUMSS > A&D, supporting the ANOVA results that academic track is significantly associated with differences in graduates' overall career satisfaction.

B. Do students with high internship experience achieve significantly higher starting salaries?

To validate our findings from EDA Question 4, we test whether completing 3 or more internships results in a significantly higher starting salary.

- H_0 **(Null Hypothesis):** The mean starting salary is equal between students with high internship experience (≥ 3) and low/moderate experience (< 3).
- H_a **(Alternative Hypothesis):** The mean starting salary is significantly higher for students with high internship experience (≥ 3)[cite: 4].
- **Significance Level (α):** 0.05[cite: 4]

In [238…

```python
# 1. Preprocessing: Split dataset into two internship groups[cite: 4]
high_intern_salary = ecs_df.loc[ecs_df["Internships_Completed"] >= 3, "Starting_Salary"]
low_intern_salary = ecs_df.loc[ecs_df["Internships_Completed"] < 3, "Starting_Salary"]

# 2. Assumption Check: Test for Normality (Shapiro-Wilk) and Equal Variance (Levene)[cite: 4]
stat_high, p_high = shapiro(high_intern_salary)
stat_low, p_low = shapiro(low_intern_salary)
stat_lev, p_lev = levene(high_intern_salary, low_intern_salary)

print("--- Assumption Checks ---")
print(f"Shapiro-Wilk (High Exp): Statistic={stat_high:.4f}, p-value={p_high:.4e}")
print(f"Shapiro-Wilk (Low Exp):  Statistic={stat_low:.4f}, p-value={p_low:.4e}")
print(f"Levene's Variance Test:  Statistic={stat_lev:.4f}, p-value={p_lev:.4e}")
```

```
--- Assumption Checks ---
Shapiro-Wilk (High Exp): Statistic=0.8931, p-value=3.5507e-11
Shapiro-Wilk (Low Exp):  Statistic=0.9424, p-value=7.9713e-07
Levene's Variance Test:  Statistic=29.8118, p-value=8.3974e-08
```

Justification for Welch's Independent t-Test

While Shapiro-Wilk p-values indicate some deviation from normality, parametric t-tests remain robust for large sample sizes ($n > 30$) due to the Central Limit Theorem[cite: 4].

Furthermore, Levene's test evaluates to $p < 0.05$, indicating unequal variances between the two groups[cite: 4]. To account for this without violating assumptions, we appropriately apply **Welch's Two-Sample t-Test** (`equal_var=False`), which adjusts for unequal variance[cite: 4].

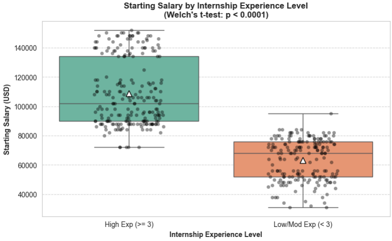
In [239…

```python
# 3. Conduct Welch's Independent Two-Sample t-Test (One-tailed: greater)[cite: 4]
t_stat, p_val = ttest_ind(high_intern_salary, low_intern_salary, equal_var=False, alternative="greater")

print("--- Welch's Independent t-Test Results ---")
print(f"t-Statistic: {t_stat:.4f}")
print(f"p-value:     {p_val:.4e}")
print(f"Significance Level (alpha): 0.05")
```

```
--- Welch's Independent t-Test Results ---
t-Statistic: 24.5372
p-value:     3.2692e-79
Significance Level (alpha): 0.05
```

In [240…

```python
# 1. Create a temporary column for clean plot labeling
ecs_df["Internship_Group"] = ecs_df["Internships_Completed"].apply(
    lambda x: "High Exp (>= 3)" if x >= 3 else "Low/Mod Exp (< 3)"
)

# 2. Set up the figure (Counts as 1 Figure toward your 15-figure limit!)
plt.figure(figsize=(8, 5))

# 3. Create Boxplot combined with Stripplot (Updated to remove Seaborn warning!)
sns.boxplot(
    data=ecs_df,
    x="Internship_Group",
    y="Starting_Salary",
    hue="Internship_Group",  # Added to fix the warning
    palette="Set2",
    legend=False,            # Added to prevent duplicate legends
    showmeans=True,  # Shows the mean as a triangle marker
    meanprops={"marker":"^", "markerfacecolor":"white", "markeredgecolor":"black", "markersize":"8"}
)
sns.stripplot(
    data=ecs_df,
    x="Internship_Group",
    y="Starting_Salary",
    color="black",
    alpha=0.4,
    jitter=0.2
)

# 4. Customize labels and titles
plt.title("Starting Salary by Internship Experience Level\n(Welch's t-test: p < 0.0001)", fontsize=12, fontweight="bold")
plt.xlabel("Internship Experience Level", fontweight="bold")
plt.ylabel("Starting Salary (USD)", fontweight="bold")
plt.grid(axis="y", linestyle="--", alpha=0.7)

# Clean up temporary column
ecs_df.drop(columns=["Internship_Group"], inplace=True)

plt.tight_layout()
plt.show()
```



Statistical Conclusion

Welch's t-test resulted in an extremely small p-value ($p < 0.0001$), which is well below our significance threshold of $\alpha = 0.05$[cite: 4]. Therefore, we **reject the null hypothesis**[cite: 4].

We conclude with strong statistical significance that students who complete 3 or more internships secure a higher mean starting salary compared to those with fewer internships[cite: 4].

C. Is there a significant monotonic association between starting salary and job satisfaction as well as work-life balance and job satisfaction?

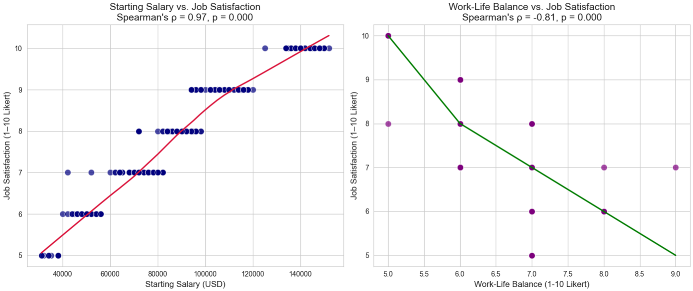
To check if starting salary and job satisfaction, as well as work-life balance and job satisfaction, are truly related, we will run the data through a Spearman's correlation to check if there exists a monotonic association between them. Spearman is preferred over Pearson as we are dealing with ordinal, discrete data while Pearson prefers continuous data.

- H_0 (Null Hypothesis): ρ = 0 (There is no monotonic association between the two variables).
- H_a (Alternative Hypothesis): ρ ≠ 0 (There is a monotonic association between the two variables)
- Decision Rule: At a 95% confidence level (α = 0.05), we reject H_0 if p ≤ 0.05.

In [241…

```python
# create sub-dataset for testingg
df_eda3test = ecs_df[['Starting_Salary', 'Career_Satisfaction', 'Work_Life_Balance']].dropna()
print(f"Loaded {len(df_eda3test)} complete records.")

# Starting Salary vs Career Satisfaction
rho3 = df_eda3test['Starting_Salary'].corr(df_eda3test['Career_Satisfaction'], method='spearman')
print(f"SS vs CS Spearman's rank correlation coefficient (ρ) = {rho3:.4f}")

# Work-Life Balance vs Career Satisfaction
rho4 = df_eda3test['Work_Life_Balance'].corr(df_eda3test['Career_Satisfaction'], method='spearman')
print(f"WLB vs CS Spearman's rank correlation coefficient (ρ) = {rho4:.4f}")

# Set up the permutation test to check if the rho values are legit
observed_rho = rho3
observed_rho2 = rho4
n_permutations = 10000
count_extreme = 0
count_extreme2 = 0

for _ in range(n_permutations):
    # Shuffle the satisfaction ranks (breaks any real association)
    shuffled_sat = df_eda3test['Career_Satisfaction'].sample(frac=1, ignore_index=True)
    perm_rho = df_eda3test['Starting_Salary'].corr(shuffled_sat, method='spearman')
    perm_rho2 = df_eda3test['Work_Life_Balance'].corr(shuffled_sat, method='spearman')

    # Count if the shuffled correlation is as strong or stronger than observed
    if abs(perm_rho) >= abs(observed_rho):
        count_extreme += 1
    if abs(perm_rho2) >= abs(observed_rho2):
        count_extreme2 += 1

# Two-tailed p-value (add 1 to numerator and denominator to avoid 0)
p_value = (count_extreme + 1) / (n_permutations + 1)
p_value2 = (count_extreme2 + 1) / (n_permutations + 1)
print(f"SS vs CS p-value (permutation test) = {p_value:.4f}")
print(f"WLB vs CS p-value (permutation test) = {p_value2:.4f}")

# Set up scatterplot
sns.set_style("whitegrid")
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

# SS vs CS scatterplot
sns.scatterplot(data=df_eda3test, x='Starting_Salary', y='Career_Satisfaction', s=80, color='navy', alpha=0.7, ax=ax1)
sns.regplot(data=df_eda3test, x='Starting_Salary', y='Career_Satisfaction', lowess=True, scatter=False, color='crimson', line_kws={'linewidth': 2}, ax=ax1)

ax1.set_title(f"Starting Salary vs. Job Satisfaction\nSpearman's ρ = {rho3:.2f}, p = {p_value:.3f}", fontsize=14)
ax1.set_xlabel("Starting Salary (USD)", fontsize=12)
ax1.set_ylabel("Job Satisfaction (1-10 Likert)", fontsize=12)

# WLB vs CS scatterplot
sns.scatterplot(data=df_eda3test, x='Work_Life_Balance', y='Career_Satisfaction', s=80, color='purple', alpha=0.7, ax=ax2)
sns.regplot(data=df_eda3test, x='Work_Life_Balance', y='Career_Satisfaction', lowess=True, scatter=False, color='green', line_kws={'linewidth': 2}, ax=ax2)

ax2.set_title(f"Work-Life Balance vs. Job Satisfaction\nSpearman's ρ = {rho4:.2f}, p = {p_value2:.3f}", fontsize=14)
ax2.set_xlabel("Work-Life Balance (1-10 Likert)", fontsize=12)
ax2.set_ylabel("Job Satisfaction (1-10 Likert)", fontsize=12)

# Print scatterplots
plt.tight_layout()
plt.show()
```

```
Loaded 400 complete records.
SS vs CS Spearman's rank correlation coefficient (ρ) = 0.9700
WLB vs CS Spearman's rank correlation coefficient (ρ) = -0.8080
SS vs CS p-value (permutation test) = 0.0001
WLB vs CS p-value (permutation test) = 0.0001
```



The Spearman's correlation's results reveals a very strong, positive monotonic association between Starting Salary and Job Satisfaction, with ρ at 0.970 and p at 0.0001. This indicates that higher starting salaries are consistently associated with higher levels of career satisfaction in the sample, and this relationship is highly unlikely to be due to random sampling error (p < 0.001).

The Spearman's correlation's results also reveal a very strong, negative monotonic association between Work-Life Balance and Job Satisfaction, with ρ at -0.81 and p at 0.0001. This indicates that higher work-life balance scores are consistently associated with lower levels of career satisfaction in the sample, and this relationship is highly unlikely to be due to random sampling error (p < 0.001).

Statistical Conclusion

Welch's t-test resulted in an extremely small p-value ($p < 0.0001$), which is well below our significance threshold of $\alpha = 0.05$[cite: 4]. Therefore, we **reject the null hypothesis**[cite: 4].

We conclude with strong statistical significance that students who complete 3 or more internships secure a higher mean starting salary compared to those with fewer internships[cite: 4].

D. Is a student's academic track associated with their current job level?

This tests whether the distribution of Current_Job_Level (Entry, Mid, Senior, Executive) differs across Track (STEM, ABM, HUMSS, A&D).

- **H_0 (Null Hypothesis)**: Track and Current_Job_Level are independent. The distribution of job levels is the same across academic tracks.
- **H_a (Alternative Hypothesis)**: Track and Current_Job_Level are associated. The distribution of job levels differs across academic tracks.
- Significance level: α = 0.05

Through the use of the Chi-Square Test of Independence to see the relationship between academic track, & current job level:

In [242…

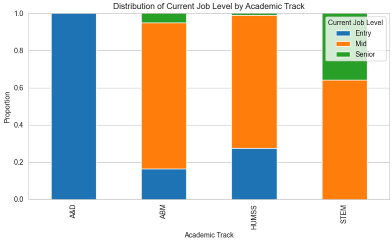
```
# Contingency table
contingency_table = pd.crosstab(ecs_df['Track'], ecs_df['Current_Job_Level'])

print("Contingency Table:")
print(contingency_table)
print()

# Chi-square test of independence
chi2, p, dof, expected = chi2_contingency(contingency_table)

print("Chi-square Test Results:")
print(f"Chi-square statistic: {chi2:.4f}")
print(f"Degrees of freedom: {dof}")
print(f"p-value: {p:.4e}")

# Expected frequencies
expected_df = pd.DataFrame(
    expected,
    index=contingency_table.index,
    columns=contingency_table.columns
)

print("Expected Frequencies:")
print(expected_df.round(2))

# Assumption check
print("Minimum expected frequency:", expected.min())
```

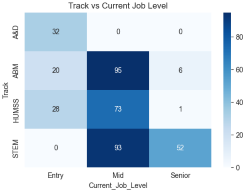
```
Contingency Table:
Current_Job_Level  Entry  Mid  Senior
Track
A&D                   32    0       0
ABM                   20   95       6
HUMSS                 28   73       1
STEM                   0   93      52

Chi-square Test Results:
Chi-square statistic: 229.2662
Degrees of freedom: 6
p-value: 1.0981e-46
Expected Frequencies:
Current_Job_Level  Entry   Mid  Senior
Track
A&D                 6.40 20.88    4.72
ABM                24.20 78.95   17.85
HUMSS              20.40 66.56   15.04
STEM               29.00 94.61   21.39
Minimum expected frequency: 4.72
```

To visualize the results of the Chi-Square Test:

In [243…

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Stacked Bar Chart
prop_table = pd.crosstab(
    ecs_df["Track"],
    ecs_df["Current_Job_Level"],
    normalize="index"
)

prop_table.plot(
    kind="bar",
    stacked=True,
    figsize=(8,5)
)

plt.title("Distribution of Current Job Level by Academic Track")
plt.xlabel("Academic Track")
plt.ylabel("Proportion")
plt.legend(title="Current Job Level")
plt.tight_layout()
plt.show()

# Heatmap
contingency = pd.crosstab(
    ecs_df["Track"],
    ecs_df["Current_Job_Level"]
)

plt.figure(figsize=(6,4))
sns.heatmap(contingency,
            annot=True,
            fmt="d",
            cmap="Blues")

plt.title("Track vs Current Job Level")
plt.show()
```





The stacked bar chart and heatmap visually support the results of the Chi-square Test of Independence, illustrating an association between academic track and current job level. STEM graduates are concentrated in mid- and senior-level positions, with no respondents in entry-level roles. On the other hand, all respondents from the A&D track are in entry-level positions. ABM and HUMSS graduates are predominantly employed in mid-level positions, with relatively few reaching senior-level roles.

The distribution across the tracks are consistent with the statistically significant association identified by the Chi-square test.

Statistical conclusion

The Chi-square Test of Independence produced the following results:

- $\chi^2 = 229.2662$
- df = 6
- p-value = 1.0981×10^{-46}

Since the p-value is much smaller than the significance level of 0.05, we reject the null hypothesis. Therefore, there is sufficient statistical evidence to conclude that academic track and current job level are significantly associated. This indicates that the distribution of current job levels differs across academic tracks and is unlikely to have occurred by chance alone.

The stacked bar chart and heatmap further support this finding by showing the distinct, job-level distributions among the different academic tracks. For example, STEM graduates are concentrated in mid- and senior-level positions, with no respondents in entry-level roles, whereas all A&D graduates are in entry-level positions. These observed patterns are consistent with the statistically significant association identified by the Chi-square test.

E. Are the Fields of Study associated with the Skill & Extracurricular columns?

EDA 5 attempted to find if the Fields of study influence the number of skills & extracurriculars a student takes. An intial conlcusion was found that STEM aligned fields tend to take more skills & extracurriculars, with ABM fields coming second. To determine if the results from EDA 5 are statistically significant, perform an ANOVA test to verify/reject the initial observation.

Since There are five columns under skills & extracurriculars, perform ANOVA testing on each of the five columns.

- H_0 (Null Hypothesis): There is no difference between the fields of study in any of the skills & extracurriculars columns.
- H_a (Alternative Hypothesis): At least one field of study differes from the rest in terms of skills & extracurriculars.
- Significance Level (α): 0.05

In [244…

```
# Skills and Extracurriculars columns
skill_cols = [
    'Internships_Completed',
    'Projects_Completed',
    'Certifications',
    'Soft_Skills_Score',
    'Networking_Score'
]

alpha = 0.5
results = []

for col in skill_cols:
    groups = [
        group[col].dropna().values
        for name, group in ecs_df.groupby('Field_of_Study')
    ]

    f_stat, p_value = f_oneway(*groups)

    sig = p_value < alpha
    conclusion = "Reject H0 (significant)" if sig else "Fail to reject H0 (not significant)"

    results.append({
        'Column': col,
        'F-statistic': round(f_stat, 4),
        'p-value': round(p_value, 4),
        'Significant (a=0.05)': sig,
        'Conclusion': conclusion
    })

results_df = pd.DataFrame(results)
results_df
```

Out[244…

	Column	F-statistic	p-value	Significant (a=0.05)	Conclusion
0	Internships_Completed	46.55	0.00	True	Reject H0 (significant)
1	Projects_Completed	54.65	0.00	True	Reject H0 (significant)
2	Certifications	54.28	0.00	True	Reject H0 (significant)
3	Soft_Skills_Score	51.74	0.00	True	Reject H0 (significant)
4	Networking_Score	52.47	0.00	True	Reject H0 (significant)

Note: The p-value is 0.00 due to the rounding done. If the p-value results to 0, it means that the computed p-value is significantly small, rounding to 0.

After the ANOVA testing, results showed that all the tests came back as significant and we reject the null hypothesis. What this means is that there is some difference within those groups however it does not tell us which specific filed/s those are. So to further understand and the results and formulate a better conclusion we will perform post-hoc testing.

Calculate effect size using eta-squared (η^2) for ANOVA

In [245…

```
# eta squared function
def eta_squared(df, group_col, value_col):
    grand_mean = df[value_col].mean()
    ss_total = ((df[value_col] - grand_mean) ** 2).sum()

    ss_between = 0
    for name, group in df.groupby(group_col):
        group_mean = group[value_col].mean()
        n = len(group)
        ss_between += n * (group_mean - grand_mean) ** 2

    return ss_between / ss_total
```

In [246…

```
effect_sizes = {}
for var in skill_cols:
    eta_sq = eta_squared(ecs_df, 'Field_of_Study', var)
    effect_sizes[var] = eta_sq

    if eta_sq < 0.06:
        magnitude = "small"
    elif eta_sq < 0.14:
        magnitude = "medium"
    else:
        magnitude = "large"

    print(f"{var}: \tη² = {eta_sq:.4f} ({magnitude} effect, {eta_sq*100:.1f}% of variance explained by Field_of_Study)")
```

```
Internships_Completed:  η² = 0.5448 (large effect, 54.5% of variance explained by Field_of_Study)
Projects_Completed:     η² = 0.5842 (large effect, 58.4% of variance explained by Field_of_Study)
Certifications:         η² = 0.5825 (large effect, 58.3% of variance explained by Field_of_Study)
Soft_Skills_Score:      η² = 0.5708 (large effect, 57.1% of variance explained by Field_of_Study)
Networking_Score:       η² = 0.5743 (large effect, 57.4% of variance explained by Field_of_Study)
```

The results from the effect size show that there are large effects within between the fields of study for each skill & extracurricular. the largest observed variance was with Projects_Completed resulting in 58.4% of variance due to Field_of_Study.

Next we perform a post-hoc pairwise test using Tukey HSD. The results from this show which specific field pairs are significantly different/not different.

In [247…

```python
tk_res = {}
for skill in skill_cols:
    tk = pairwise_tukeyhsd( ecs_df[skill], ecs_df['Field_of_Study'], alpha )
    tk_res[skill] = tk

    print(f"{skill}")
    print(tk.summary())
    print("\n")
```

```
Internships_Completed
          Multiple Comparison of Means - Tukey HSD, FWER=0.50
========================================================================
     group1           group2      meandiff p-adj   lower   upper  reject
------------------------------------------------------------------------
            Arts         Business   1.6042    0.0  1.2636  1.9447   True
            Arts Computer Science   2.9656    0.0  2.6013  3.3299   True
            Arts        Education   0.6458 0.9276  -0.322  1.6136  False
            Arts      Engineering   2.1573    0.0  1.8044  2.5103   True
            Arts          Finance   1.9125    0.0  1.1417  2.6833   True
            Arts              Law   1.6989    0.0  1.3265  2.0712   True
            Arts        Marketing   1.8352    0.0  1.4628  2.2076   True
            Arts         Medicine   2.8839    0.0  2.4919   3.276   True
            Arts          Nursing   1.3125 0.0991  0.3447  2.2803   True
            Arts       Psychology   1.2216    0.0  0.8652  1.5779   True
        Business Computer Science   1.3614    0.0  1.0646  1.6582   True
        Business        Education  -0.9583  0.477 -1.9028 -0.0139   True
        Business      Engineering   0.5532  0.001  0.2704   0.836   True
        Business          Finance   0.3083 0.9979 -0.4329  1.0496  False
        Business              Law   0.0947 0.9998  -0.212  0.4014  False
        Business        Marketing   0.2311 0.8529 -0.0756  0.5378  False
        Business         Medicine   1.2798    0.0  0.9495    1.61   True
        Business          Nursing  -0.2917 0.9998 -1.2361  0.6528  False
        Business       Psychology  -0.3826 0.1134 -0.6696 -0.0955   True
Computer Science        Education  -2.3197    0.0  -3.273 -1.3664   True
Computer Science      Engineering  -0.8082    0.0 -1.1192 -0.4972   True
Computer Science          Finance  -1.0531 0.0764 -1.8055 -0.3006   True
Computer Science              Law  -1.2667    0.0 -1.5996 -0.9338   True
Computer Science        Marketing  -1.1303    0.0 -1.4632 -0.7974   True
Computer Science         Medicine  -0.0816    1.0 -0.4364  0.2731  False
Computer Science          Nursing  -1.6531 0.0069 -2.6064 -0.6998   True
Computer Science       Psychology   -1.744    0.0 -2.0588 -1.4291   True
       Education      Engineering   1.5115 0.0207  0.5625  2.4605   True
       Education          Finance   1.2667 0.3757  0.0961  2.4372   True
       Education              Law    1.053 0.3491  0.0966  2.0094   True
       Education        Marketing   1.1894 0.1833   0.233  2.1458   True
       Education         Medicine   2.2381    0.0  1.2739  3.2023   True
       Education          Nursing   0.6667 0.9891  -0.642  1.9754  False
       Education       Psychology   0.5758 0.9615 -0.3745   1.526  False
     Engineering          Finance  -0.2448 0.9997 -0.9919  0.5022  False
     Engineering              Law  -0.4585 0.0628 -0.7789  -0.138   True
     Engineering        Marketing  -0.3221 0.4918 -0.6425 -0.0017   True
     Engineering         Medicine   0.7266 0.0002  0.3835  1.0697   True
     Engineering          Nursing  -0.8448 0.6725 -1.7938  0.1042  False
     Engineering       Psychology  -0.9357    0.0 -1.2374 -0.6341   True
         Finance              Law  -0.2136 0.9999 -0.9701  0.5428  False
         Finance        Marketing  -0.0773    1.0 -0.8337  0.6792  False
         Finance         Medicine   0.9714 0.1619  0.2051  1.7377   True
         Finance          Nursing     -0.6 0.9885 -1.7705  0.5705  False
         Finance       Psychology  -0.6909 0.6221 -1.4396  0.0578  False
             Law        Marketing   0.1364 0.9985 -0.2054  0.4781  False
             Law         Medicine   1.1851    0.0   0.822  1.5481   True
             Law          Nursing  -0.3864 0.9983 -1.3428    0.57  False
             Law       Psychology  -0.4773  0.048 -0.8015 -0.1531   True
       Marketing         Medicine   1.0487    0.0  0.6857  1.4117   True
       Marketing          Nursing  -0.5227 0.9815 -1.4791  0.4337  False
       Marketing       Psychology  -0.6136 0.0018 -0.9378 -0.2895   True
        Medicine          Nursing  -1.5714 0.0157 -2.5357 -0.6072   True
        Medicine       Psychology  -1.6623    0.0 -2.0089 -1.3158   True
         Nursing       Psychology  -0.0909    1.0 -1.0412  0.8594  False
------------------------------------------------------------------------


Projects_Completed
          Multiple Comparison of Means - Tukey HSD, FWER=0.50
========================================================================
     group1           group2      meandiff p-adj   lower   upper  reject
------------------------------------------------------------------------
            Arts         Business   3.3819    0.0  2.7989   3.965   True
            Arts Computer Science   5.6773    0.0  5.0536   6.301   True
            Arts        Education   1.5208 0.6299 -0.1361  3.1777  False
            Arts      Engineering   3.9461    0.0  3.3419  4.5504   True
            Arts          Finance   3.5875    0.0  2.2679  4.9071   True
            Arts              Law   2.9602    0.0  2.3227  3.5978   True
            Arts        Marketing   3.3011    0.0  2.6636  3.9387   True
            Arts         Medicine   5.2446    0.0  4.5735  5.9158   True
            Arts          Nursing   2.8542 0.0076  1.1973  4.5111   True
            Arts       Psychology    2.333    0.0  1.7229  2.9431   True
        Business Computer Science   2.2954    0.0  1.7872  2.8035   True
        Business        Education  -1.8611 0.2836 -3.4781 -0.2441   True
        Business      Engineering   0.5642 0.2662    0.08  1.0483   True
        Business          Finance   0.2056    1.0 -1.0635  1.4746  False
        Business              Law  -0.4217 0.7949 -0.9468  0.1034  False
        Business        Marketing  -0.0808    1.0 -0.6059  0.4443  False
        Business         Medicine   1.8627    0.0  1.2973  2.4281   True
        Business          Nursing  -0.5278 0.9997 -2.1447  1.0892  False
        Business       Psychology   -1.049 0.0002 -1.5404 -0.5576   True
Computer Science        Education  -4.1565    0.0 -5.7885 -2.5244   True
Computer Science      Engineering  -1.7312    0.0 -2.2636 -1.1987   True
Computer Science          Finance  -2.0898 0.0166 -3.3781 -0.8015   True
Computer Science              Law  -2.7171    0.0  -3.287 -2.1471   True
Computer Science        Marketing  -2.3762    0.0 -2.9461 -1.8062   True
Computer Science         Medicine  -0.4327 0.8926   -1.04  0.1747  False
Computer Science          Nursing  -2.8231 0.0071 -4.4552  -1.191   True
Computer Science       Psychology  -3.3443    0.0 -3.8834 -2.8053   True
       Education      Engineering   2.4253 0.0418  0.8005    4.05   True
       Education          Finance   2.0667 0.4513  0.0627  4.0707   True
       Education              Law   1.4394 0.6893  -0.198  3.0768  False
       Education        Marketing   1.7803 0.3684  0.1429  3.4177   True
       Education         Medicine   3.7238 0.0001   2.073  5.3746   True
       Education          Nursing   1.3333  0.966 -0.9072  3.5739  False
       Education       Psychology   0.8121 0.9907 -0.8148  2.4391  False
     Engineering          Finance  -0.3586 0.9999 -1.6376  0.9204  False
     Engineering              Law  -0.9859 0.0041 -1.5345 -0.4373   True
     Engineering        Marketing   -0.645  0.254 -1.1936 -0.0964   True
     Engineering         Medicine   1.2985 0.0001  0.7112  1.8859   True
     Engineering          Nursing   -1.092 0.9243 -2.7167  0.5328  False
     Engineering       Psychology  -1.6132    0.0 -2.1296 -1.0967   True
         Finance              Law  -0.6273 0.9926 -1.9223  0.6678  False
         Finance        Marketing  -0.2864    1.0 -1.5814  1.0087  False
         Finance         Medicine   1.6571 0.1658  0.3452  2.9691   True
         Finance          Nursing  -0.7333 0.9993 -2.7373  1.2707  False
         Finance       Psychology  -1.2545 0.5336 -2.5363  0.0272  False
             Law        Marketing   0.3409 0.9706 -0.2441   0.926  False
             Law         Medicine   2.2844    0.0  1.6629  2.9059   True
             Law          Nursing  -0.1061    1.0 -1.7435  1.5314  False
             Law       Psychology  -0.6273 0.3099 -1.1823 -0.0723   True
       Marketing         Medicine   1.9435    0.0   1.322   2.565   True
       Marketing          Nursing   -0.447 0.9999 -2.0844  1.1905  False
       Marketing       Psychology  -0.9682 0.0063 -1.5232 -0.4132   True
        Medicine          Nursing  -2.3905 0.0562 -4.0413 -0.7397   True
        Medicine       Psychology  -2.9117    0.0  -3.505 -2.3183   True
         Nursing       Psychology  -0.5212 0.9998 -2.1481  1.1057  False
------------------------------------------------------------------------


Certifications
          Multiple Comparison of Means - Tukey HSD, FWER=0.50
========================================================================
     group1           group2      meandiff p-adj   lower   upper  reject
------------------------------------------------------------------------
```

```
            Arts         Business   1.6181    0.0   1.228  2.0081   True
            Arts Computer Science   3.5982    0.0  3.1809  4.0155   True
            Arts        Education   0.6458 0.9707 -0.4627  1.7544  False
            Arts      Engineering   2.1746    0.0  1.7703  2.5788   True
            Arts          Finance   1.9125 0.0001  1.0297  2.7953   True
            Arts              Law   1.6989    0.0  1.2723  2.1254   True
            Arts        Marketing   1.8352    0.0  1.4087  2.2618   True
            Arts         Medicine   3.3696    0.0  2.9206  3.8187   True
            Arts          Nursing   1.3125 0.2445   0.204   2.421   True
            Arts       Psychology   1.2216    0.0  0.8134  1.6298   True
        Business Computer Science   1.9802    0.0  1.6402  2.3202   True
        Business        Education  -0.9722 0.6596  -2.054  0.1096  False
        Business      Engineering   0.5565 0.0079  0.2326  0.8804   True
        Business          Finance   0.2944 0.9996 -0.5546  1.1435  False
        Business              Law   0.0808    1.0 -0.2705  0.4321  False
        Business        Marketing   0.2172 0.9559 -0.1341  0.5685  False
        Business         Medicine   1.7516    0.0  1.3733  2.1299   True
        Business          Nursing  -0.3056 0.9999 -1.3874  0.7763  False
        Business       Psychology  -0.3965 0.2207 -0.7252 -0.0677   True
Computer Science        Education  -2.9524    0.0 -4.0443 -1.8605   True
Computer Science      Engineering  -1.4236    0.0 -1.7799 -1.0674   True
Computer Science          Finance  -1.6857  0.001 -2.5476 -0.8238   True
Computer Science              Law  -1.8994    0.0 -2.2806 -1.5181   True
Computer Science        Marketing   -1.763    0.0 -2.1443 -1.3817   True
Computer Science         Medicine  -0.2286 0.9771 -0.6349  0.1777  False
Computer Science          Nursing  -2.2857 0.0003 -3.3776 -1.1938   True
Computer Science       Psychology  -2.3766    0.0 -2.7373  -2.016   True
       Education      Engineering   1.5287 0.0732  0.4417  2.6158   True
       Education          Finance   1.2667 0.5876 -0.0741  2.6074  False
       Education              Law    1.053 0.5614 -0.0425  2.1485  False
       Education        Marketing   1.1894 0.3706  0.0939  2.2849   True
       Education         Medicine   2.7238    0.0  1.6194  3.8283   True
       Education          Nursing   0.6667 0.9963 -0.8323  2.1657  False
       Education       Psychology   0.5758 0.9855 -0.5127  1.6642  False
     Engineering          Finance  -0.2621 0.9999 -1.1178  0.5936  False
     Engineering              Law  -0.4757 0.1391 -0.8427 -0.1087   True
     Engineering        Marketing  -0.3393 0.6194 -0.7064  0.0277  False
     Engineering         Medicine   1.1951    0.0  0.8021   1.588   True
     Engineering          Nursing  -0.8621 0.8074 -1.9491  0.2249  False
     Engineering       Psychology   -0.953    0.0 -1.2985 -0.6074   True
         Finance              Law  -0.2136    1.0 -1.0801  0.6528  False
         Finance        Marketing  -0.0773    1.0 -0.9437  0.7892  False
         Finance         Medicine   1.4571 0.0124  0.5794  2.3349   True
         Finance          Nursing     -0.6 0.9961 -1.9407  0.7407  False
         Finance       Psychology  -0.6909 0.7916 -1.5484  0.1666  False
             Law        Marketing   0.1364 0.9995  -0.255  0.5278  False
             Law         Medicine   1.6708    0.0   1.255  2.0866   True
             Law          Nursing  -0.3864 0.9995 -1.4819  0.7091  False
             Law       Psychology  -0.4773 0.1475 -0.8486 -0.1059   True
       Marketing         Medicine   1.5344    0.0  1.1186  1.9502   True
       Marketing          Nursing  -0.5227 0.9934 -1.6182  0.5728  False
       Marketing       Psychology  -0.6136 0.0132  -0.985 -0.2423   True
        Medicine          Nursing  -2.0571 0.0023 -3.1616 -0.9527   True
        Medicine       Psychology  -2.1481    0.0  -2.545 -1.7511   True
         Nursing       Psychology  -0.0909    1.0 -1.1794  0.9976  False
------------------------------------------------------------------------


Soft_Skills_Score
          Multiple Comparison of Means - Tukey HSD, FWER=0.50
========================================================================
     group1           group2      meandiff p-adj   lower   upper  reject
------------------------------------------------------------------------
            Arts         Business   1.9688    0.0  1.5596  2.3779   True
            Arts Computer Science   3.7902    0.0  3.3524  4.2279   True
            Arts        Education   1.2188 0.4259  0.0558  2.3817   True
            Arts      Engineering   2.3222    0.0  1.8981  2.7463   True
            Arts          Finance   2.2188    0.0  1.2926  3.1449   True
            Arts              Law   1.8778    0.0  1.4304  2.3253   True
            Arts        Marketing   2.0369    0.0  1.5895  2.4844   True
            Arts         Medicine   3.6188    0.0  3.1477  4.0898   True
            Arts          Nursing   1.5521 0.1122  0.3892   2.715   True
            Arts       Psychology   1.4551    0.0  1.0269  1.8833   True
        Business Computer Science   1.8214    0.0  1.4648  2.1781   True
        Business        Education    -0.75 0.9319 -1.8849  0.3849  False
        Business      Engineering   0.3534 0.4378  0.0136  0.6933   True
        Business          Finance     0.25 0.9999 -0.6407  1.1407  False
        Business              Law  -0.0909    1.0 -0.4594  0.2776  False
        Business        Marketing   0.0682    1.0 -0.3004  0.4367  False
        Business         Medicine     1.65    0.0  1.2531  2.0469   True
        Business          Nursing  -0.4167 0.9993 -1.5515  0.7182  False
        Business       Psychology  -0.5136 0.0428 -0.8585 -0.1687   True
Computer Science        Education  -2.5714 0.0001 -3.7169 -1.4259   True
Computer Science      Engineering   -1.468    0.0 -1.8417 -1.0943   True
Computer Science          Finance  -1.5714 0.0067 -2.4756 -0.6672   True
Computer Science              Law  -1.9123    0.0 -2.3123 -1.5123   True
Computer Science        Marketing  -1.7532    0.0 -2.1532 -1.3532   True
Computer Science         Medicine  -0.1714 0.9984 -0.5977  0.2548  False
Computer Science          Nursing  -2.2381  0.001 -3.3836 -1.0926   True
Computer Science       Psychology  -2.3351    0.0 -2.7134 -1.9567   True
       Education      Engineering   1.1034 0.5512 -0.0369  2.2438  False
       Education          Finance      1.0 0.8938 -0.4065  2.4065  False
       Education              Law   0.6591 0.9738 -0.4901  1.8083  False
       Education        Marketing   0.8182  0.893 -0.3311  1.9674  False
       Education         Medicine      2.4 0.0003  1.2414  3.5586   True
       Education          Nursing   0.3333    1.0 -1.2392  1.9059  False
       Education       Psychology   0.2364    1.0 -0.9055  1.3782  False
     Engineering          Finance  -0.1034    1.0 -1.0011  0.7942  False
     Engineering              Law  -0.4444 0.2798 -0.8294 -0.0593   True
     Engineering        Marketing  -0.2853 0.8658 -0.6703  0.0998  False
     Engineering         Medicine   1.2966    0.0  0.8843  1.7088   True
     Engineering          Nursing  -0.7701  0.922 -1.9105  0.3702  False
     Engineering       Psychology  -0.8671    0.0 -1.2296 -0.5046   True
         Finance              Law  -0.3409 0.9991 -1.2498   0.568  False
         Finance        Marketing  -0.1818    1.0 -1.0908  0.7271  False
         Finance         Medicine      1.4 0.0346  0.4792  2.3208   True
         Finance          Nursing  -0.6667 0.9938 -2.0732  0.7399  False
         Finance       Psychology  -0.7636 0.7334 -1.6632   0.136  False
             Law        Marketing   0.1591 0.9988 -0.2515  0.5697  False
             Law         Medicine   1.7409    0.0  1.3047  2.1771   True
             Law          Nursing  -0.3258 0.9999  -1.475  0.8235  False
             Law       Psychology  -0.4227 0.3713 -0.8123 -0.0332   True
       Marketing         Medicine   1.5818    0.0  1.1456   2.018   True
       Marketing          Nursing  -0.4848 0.9976 -1.6341  0.6644  False
       Marketing       Psychology  -0.5818 0.0416 -0.9714 -0.1923   True
        Medicine          Nursing  -2.0667 0.0046 -3.2253  -0.908   True
        Medicine       Psychology  -2.1636    0.0 -2.5801 -1.7472   True
         Nursing       Psychology   -0.097    1.0 -1.2388  1.0449  False
------------------------------------------------------------------------


Networking_Score
          Multiple Comparison of Means - Tukey HSD, FWER=0.50
========================================================================
     group1           group2      meandiff p-adj   lower   upper  reject
------------------------------------------------------------------------
```

```
           Arts         Business     2.0    0.0  1.5913  2.4087   True
           Arts Computer Science  3.8214    0.0  3.3842  4.2587   True
           Arts        Education    1.25 0.3843  0.0884  2.4116   True
           Arts      Engineering  2.3534    0.0  1.9298  2.7771   True
           Arts          Finance    2.45    0.0  1.5249  3.3751   True
           Arts              Law  1.9091    0.0  1.4621   2.356   True
           Arts        Marketing  2.0682    0.0  1.6212  2.5151   True
           Arts         Medicine    3.65    0.0  3.1795  4.1205   True
           Arts          Nursing  1.5833 0.0951  0.4217  2.7449   True
           Arts       Psychology  1.4864    0.0  1.0586  1.9141   True
       Business Computer Science  1.8214    0.0  1.4652  2.1777   True
       Business        Education   -0.75 0.9314 -1.8836  0.3836  False
       Business      Engineering  0.3534  0.436   0.014  0.6929   True
       Business          Finance    0.45 0.9897 -0.4397  1.3397  False
       Business              Law -0.0909    1.0  -0.459  0.2772  False
       Business        Marketing  0.0682    1.0 -0.2999  0.4363  False
       Business         Medicine    1.65    0.0  1.2536  2.0464   True
       Business          Nursing -0.4167 0.9993 -1.5503  0.7169  False
       Business       Psychology -0.5136 0.0424 -0.8582 -0.1691   True
Computer Science        Education -2.5714 0.0001 -3.7156 -1.4272   True
Computer Science      Engineering  -1.468    0.0 -1.8413 -1.0947   True
Computer Science          Finance -1.3714 0.0351 -2.2746 -0.4683   True
Computer Science              Law -1.9123    0.0 -2.3119 -1.5128   True
Computer Science        Marketing -1.7532    0.0 -2.1528 -1.3537   True
Computer Science         Medicine -0.1714 0.9984 -0.5972  0.2543  False
Computer Science          Nursing -2.2381  0.001 -3.3823 -1.0939   True
Computer Science       Psychology -2.3351    0.0  -2.713 -1.9572   True
      Education      Engineering  1.1034 0.5495 -0.0356  2.2425  False
      Education          Finance     1.2 0.7259 -0.2049  2.6049  False
      Education              Law  0.6591 0.9735 -0.4888   1.807  False
      Education        Marketing  0.8182 0.8923 -0.3298  1.9661  False
      Education         Medicine     2.4 0.0003  1.2427  3.5573   True
      Education          Nursing  0.3333    1.0 -1.2374  1.9041  False
      Education       Psychology  0.2364    1.0 -0.9042  1.3769  False
    Engineering          Finance  0.0966    1.0 -0.8001  0.9932  False
    Engineering              Law -0.4444 0.2782  -0.829 -0.0598   True
    Engineering        Marketing -0.2853 0.8649 -0.6699  0.0993  False
    Engineering         Medicine  1.2966    0.0  0.8848  1.7083   True
    Engineering          Nursing -0.7701 0.9215 -1.9092  0.3689  False
    Engineering       Psychology -0.8671    0.0 -1.2292  -0.505   True
        Finance              Law -0.5409 0.9657 -1.4488   0.367  False
        Finance        Marketing -0.3818 0.9977 -1.2897  0.5261  False
        Finance         Medicine     1.2 0.1327  0.2803  2.1197   True
        Finance          Nursing -0.8667 0.9565 -2.2716  0.5383  False
        Finance       Psychology -0.9636 0.3898 -1.8622  -0.065   True
            Law        Marketing  0.1591 0.9988 -0.2511  0.5692  False
            Law         Medicine  1.7409    0.0  1.3052  2.1766   True
            Law          Nursing -0.3258 0.9999 -1.4737  0.8222  False
            Law       Psychology -0.4227 0.3696 -0.8118 -0.0336   True
      Marketing         Medicine  1.5818    0.0  1.1461  2.0175   True
      Marketing          Nursing -0.4848 0.9976 -1.6328  0.6631  False
      Marketing       Psychology -0.5818 0.0411 -0.9709 -0.1927   True
       Medicine          Nursing -2.0667 0.0045  -3.224 -0.9094   True
       Medicine       Psychology -2.1636    0.0 -2.5796 -1.7477   True
        Nursing       Psychology  -0.097    1.0 -1.2376  1.0436  False
---------------------------------------------------------------------
```

Apply multiple comparison correction Bonferroni (α = 0.01)

In [248…

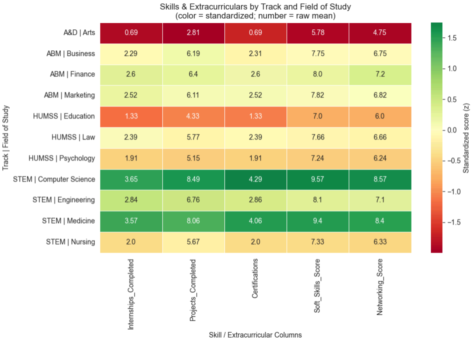
```python
alpha = 0.05
bonferroni_alpha = alpha / len(skill_cols)

for col in skill_cols:
    groups = [g[col].dropna().values for _, g in ecs_df.groupby('Field_of_Study')]
    f_stat, p_value = f_oneway(*groups)
    still_significant = p_value < bonferroni_alpha
    print(f"{col}: \tp = {p_value:.2e} -> "
          f"{'STILL significant' if still_significant else 'NOT significant'} "
         )
```

```
Internships_Completed:  p = 1.88e-60 -> STILL significant
Projects_Completed:     p = 5.63e-68 -> STILL significant
Certifications:         p = 1.20e-67 -> STILL significant
Soft_Skills_Score:      p = 2.38e-65 -> STILL significant
Networking_Score:       p = 5.09e-66 -> STILL significant
```

Results from the Bonferroni correction showed that all tests were still statistically significant even at an adjusted threshold of alpha = 0.01, indicating the current threshold still holds and proves that the test results are still significant.

In [249…

```python
# Plot heatmap for visual presentation
# Standardize (z-score) each column so variables on different scales are comparable
# Use eda4b var from EDA 5 | var is called eda4 but its from eda 5
eda4_indexed = eda4.set_index(['Track', 'Field_of_Study'])              # Re index the DF to show the tracks for each educ_field in the heatmap
eda4_z = (eda4_indexed - eda4_indexed.mean()) / eda4_indexed.std()      # Z score
eda4_sorted = eda4_indexed.sort_index(level='Track')                    # Sort by track
eda4_z_sorted = eda4_z.reindex(eda4_sorted.index)                       # Sort z scores via index from prev

row_labels = [f"{track} | {field}" for track, field in eda4_sorted.index]

plt.figure(figsize=(10, 7))
sns.heatmap(
    eda4_z_sorted.values,
    annot=eda4_sorted.round(2).values, fmt='',
    cmap='RdYlGn', center=0, linewidths=0.5,
    yticklabels=row_labels,
    xticklabels=['Internships_Completed','Projects_Completed','Certifications',
                 'Soft_Skills_Score','Networking_Score'],
    cbar_kws={'label': 'Standardized score (z)'}
)
plt.title('Skills & Extracurriculars by Track and Field of Study\n(color = standardized; number = raw mean)')
plt.ylabel('Track | Field of Study')
plt.xlabel('Skill / Extracurricular Columns')
plt.tight_layout()
plt.show()
```



	Internships_Completed	Projects_Completed	Certifications	Soft_Skills_Score	Networking_Score
A&D | Arts	0.69	2.81	0.69	5.78	4.75
ABM | Business	2.29	6.19	2.31	7.75	6.75
ABM | Finance	2.6	6.4	2.6	8.0	7.2
ABM | Marketing	2.52	6.11	2.52	7.82	6.82
HUMSS | Education	1.33	4.33	1.33	7.0	6.0
HUMSS | Law	2.39	5.77	2.39	7.66	6.66
HUMSS | Psychology	1.91	5.15	1.91	7.24	6.24
STEM | Computer Science	3.65	8.49	4.29	9.57	8.57
STEM | Engineering	2.84	6.76	2.86	8.1	7.1
STEM | Medicine	3.57	8.06	4.06	9.4	8.4
STEM | Nursing	2.0	5.67	2.0	7.33	6.33

Statistical Conclusion

The results showed that:

- The results from the all ANOVA testings showed that they are statistically significant and that there appears to be a relationship between the field of study with skill & extracurriculars. *
    - We accepted the alternative hypoyhesis, "At least one field of study differes from the rest in terms of skills & extracurriculars." for all five tests.
- The effect size is shown to be large across all Skills & Extracurriculars columns.
    - Internships_Completed: $\eta^2 = 0.5448$
    - Projects_Completed: $\eta^2 = 0.5842$
    - Certifications: $\eta^2 = 0.5825$
    - Soft_Skills_Score: $\eta^2 = 0.5708$
    - Networking_Score: $\eta^2 = 0.5743$
- Results from the multiple comparison correction Bonferroni ($\alpha = 0.01$) showed that the results are still statistically significant.
- The results from the testing back up the initial observations, meaning that there is an association between the Field of Study and Skills & Extracurriculars.
    - Can confidently say that STEM aligned fields are more likely to have more skills & extracurriculars compared to the other fields.
    - Can confidently say that A&D significantly performed the poorest across all columns, with soft skills & networking score performing the poorest.
    - STEM > ABM > HUMSS > A&D

7) Summary, Conclusions, and Recommendations

The first exploratory data analysis examined how starting salaries differ among the four academic tracks: STEM, ABM, HUMSS, and A&D. The results revealed a clear pattern in the distribution of starting salaries, with STEM graduates having the highest median starting salary and A&D graduates having the lowest. To determine whether these observed differences were statistically significant, a One-Way ANOVA was performed.

The analysis produced an F-statistic of 116.07 and a p-value of approximately 6.10×10^{-54}, which is well below the significance level of 0.05. This indicates that the mean starting salary differs significantly among the four academic tracks. Career satisfaction was also analyzed across the same academic tracks, yielding a similar pattern. The results showed that the tracks associated with higher starting salaries also reported higher levels of career satisfaction. Specifically, STEM graduates ranked highest in both starting salary and career satisfaction, followed by ABM, HUMSS, and A&D. These findings suggest that academic track is associated with multiple aspects of early career success, including both financial outcomes and overall career satisfaction.

Conclusion

The study aimed to identify the factors that contribute to higher starting salaries and overall career success among students. Based on the exploratory data analysis, clustering, and statistical inference, the results showed that both academic background and practical experience play important roles in career.

The exploratory data analysis found that STEM graduates generally received the highest starting salaries, while students with more internships, projects, certifications, networking, and soft skills also tended to have better career outcomes. Among these factors, certifications showed the strongest relationship with starting salary, suggesting that earning industry certifications may provide an advantage.

The statistical tests further supported these findings. The One-way ANOVA showed significant differences in starting salary and career satisfaction across academic tracks, while the Welch's t-test found that students with more internships earned significantly higher starting salaries. The Spearman correlation also showed a strong positive relationship between starting salary and career satisfaction. In addition, the Chi-square Test of Independence showed that academic track is significantly associated with current job level, and the One-way ANOVA on certification groups confirmed that students with more certifications tend to earn higher starting salaries.

Overall, the findings suggest that combining a strong academic background with practical experiences such as internships, projects, and certifications is associated with better career outcomes, including higher starting salaries and career satisfaction.

Recommendations

More information could be retrieved regarding the dataset. Origin, locality, data collection method, and other data collection details are missing from the source of the dataset. These information could be usefull in further undertanding the dataset.

Based on the results of this study, students are encouraged to gain more practical experience while still in school by participating in internships, completing projects, and earning relevant certifications. These experiences were consistently associated with better career outcomes throughout the analysis.

Universities may also consider providing more opportunities for students to earn industry certifications, participate in internship programs, and build professional networks through career fairs and partnerships with companies.

The dataset is very large and the group believes it still has a lot of potential for further exploration, analysis, and testing to be done. What the group has conducted and produces is only a portion of the potential research that can be done on this dataset.

Lastly, future researchers may expand this study by using larger and more diverse datasets or by including other factors that may influence career success, such as work experience, location, or industry. This can help provide a more complete understanding of the factors that contribute to career success.

8) References & Disclosures

https://www.socscistatistics.com/tutorials/tukey-hsd/

https://online.stat.psu.edu/stat502/lesson/2/2.3

https://www.statology.org/tukey-test-python/